

**CAFFEINE & CADAVER**

MBBS BIOCHEMISTRY NOTES SERIES

PYQ 2010–2025

MBBS BIOCHEMISTRY · MODULE 01

Enzyme Inhibition

Kinetics · Inhibitors · Clinical Applications

SUBJECT**Biochemistry — Paper 1****EXAM RELEVANCE****Core Curriculum Theory & Viva****PYQ FREQUENCY****Standard University Exam High-Yield**

★ APPEARED IN MBBS – YEARS

2010-S

2013

2013-S

2014-S

2015

2016

2017

2018

2019

2019-S

2021

2025

**00: LAST-MINUTE REVISION CONSOLE (READ THIS FIRST!)****Kinetics Quick Formulas**

- ✦ **Competitive:** $K_m \uparrow$ (apparent) | V_{max} unchanged
- ✦ **Non-Competitive:** K_m unchanged | $V_{max} \downarrow$
- ✦ **Uncompetitive:** $K_m \downarrow$ | $V_{max} \downarrow$

High-Yield Mechanisms

- ✦ **Suicide:** Allopurinol $\rightarrow$ metabolized to Alloxanthine $\rightarrow$ kills Xanthine Oxidase.
- ✦ **MI Markers:** LDH-1 > LDH-2 (flipped) & CK-MB > 6%.
- ✦ **Aspirin:** Irreversibly acetylates active-site Serine of COX.

**EXAM PRIORITY MAP**

- **TIER 1 (Must Do):** Competitive Kinetics, Comparison Table, K_m/V_{max} , LDH/CK, Drugs
- **TIER 2 (Important):** Non-Competitive Kinetics, Suicide, Essay Templates
- **TIER 3 (Viva):** The Big Picture, Mnemonics, Rapid Fire

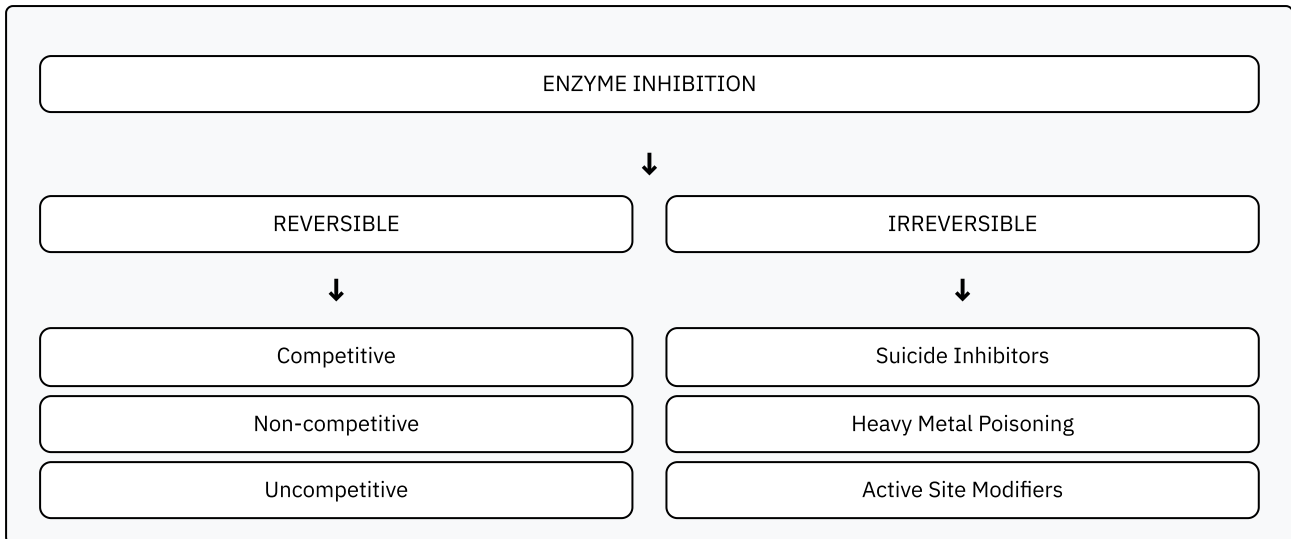
01 The Big Picture ● TIER 3**CLINICAL SIGNIFICANCE**

- ✦ **Basis of Medicine:** Over 50% of clinically prescribed drugs function through enzyme inhibition.
- ✦ **Therapeutic Leverage:** Inhibiting target enzymes stops bacterial walls (Penicillin), lowers cholesterol (Statins), or halts cancer cells (Methotrexate).
- ✦ **Kinetics Rule:** Knowing how an inhibitor alters K_m and V_{max} determines clinical dosing and how to overcome poisoning or drug resistance.

02 Types of Inhibition ● TIER 2



CLASSIFICATION TREE



- ◆ **Reversible:** Non-covalent, transient binding; enzyme recovers fully when inhibitor is cleared.
- ◆ **Irreversible:** Covalent or extremely tight binding; permanently deactivates enzyme; requires synthesis of new enzyme molecules.

03 Competitive Inhibition ● TIER 1



COMPETITIVE REVISION CARD

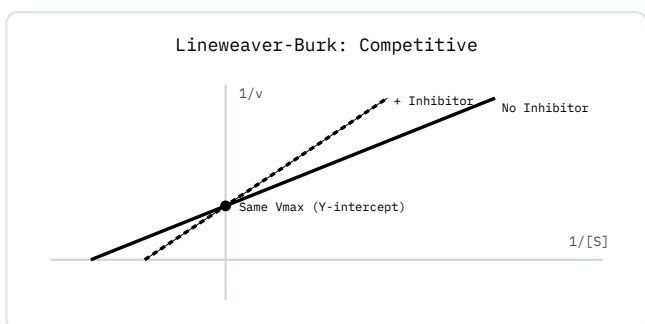
Binding Site Active Site (competes with substrate)

Apparent K_m **INCREASES** (requires more [S] to reach $\frac{1}{2} V_{max}$)

V_{max} **UNCHANGED** (high [S] fully overcomes inhibition)

● MEMORY MNEMONIC

"Competitive Kills K_m , Meets at Y"



- ◆ **Structural Similarity:** Inhibitor mimics substrate molecular shape and structure, bidding for the exact same active catalytic site.
- ◆ **Clinical Rescue:** Since binding is non-covalent, raising substrate concentration physically displaces the inhibitor, restoring full reaction velocity.
- ◆ **Classic Examples:** Statins (HMG-CoA reductase), Methotrexate (DHFR), Sulfanilamide (PABA analog).

04 Non-Competitive Inhibition ● TIER 2

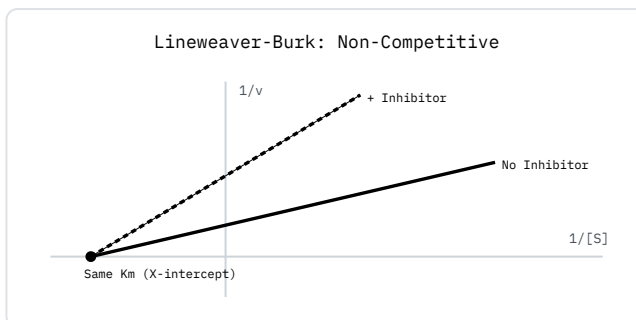


NON-COMPETITIVE REVISION CARD

Binding Site	Allosteric Site (binds free E AND ES complex)
Apparent Km	UNCHANGED (substrate binding is unaffected)
Vmax	DECREASES (reduces functional enzyme concentration)

● MEMORY MNEMONIC

"Non-Comp Nails Vmax, Meets at X"



- ◆ **Insurmountable:** Since inhibitor binds away from active site, increasing substrate concentration has zero effect and cannot rescue enzyme activity.
- ◆ **Conformational Change:** Alters overall tertiary catalytic shape, disabling enzyme conversion of ES complex to E + P.
- ◆ **Classic Examples:** Heavy metal toxicity (Hg, Pb, As binding to sulfhydryl groups), Cyanide (Complex IV blocker).

05 Master Comparison ● TIER 1



KINETICS MASTER GRID

FEATURE	COMPETITIVE	NON-COMPETITIVE	UNCOMPETITIVE
BINDING SITE	Active site (E only)	Allosteric (E & ES)	ES complex only
EFFECT ON KM	↑ Increases (apparent)	No change	↓ Decreases (apparent)
EFFECT ON VMAX	No change	↓ Decreases	↓ Decreases
OVERCOME BY ↑ [S]	YES ✓	NO X	NO X
L-B INTERCEPT	Y-axis (same Vmax)	X-axis (same Km)	Parallel lines (no intersect)
KEY EXAMPLE	Statins, Methotrexate	Heavy metals, Cyanide	Lithium

06 Suicide Inhibitors ● TIER 2



SUICIDE & IRREVERSIBLE MECHANISMS

Prodrug Analog → Binds Active Site → Enzymatic Activation → Permanent Covalent Death ☠️

Drug / Toxin	Enzyme Target	Clinical Significance
Allopurinol	Xanthine Oxidase	Gout (metabolized to oxypurinol which kills XO)
Aspirin	Cyclooxygenase (COX)	Acetylation of Serine prevents thrombosis
Penicillin	Glycopeptide Transpeptidase	Bacterial cell wall death
5-Fluorouracil	Thymidylate Synthase	Oncology (blocks DNA synthesis)

- ✦ **Active Participation:** Unlike normal inhibitors, the target enzyme actively converts this inactive substrate analog into its own toxic, covalent assassin.
- ✦ **Mnemonic: "Suicide Inhibitors: APPLE"** (Allopurinol, Penicillin, Plavix/Aspirin, L-DOPA analog, Eflornithine).

07 Km & Vmax Core ● TIER 1



MICHAELIS-MENTEN & AFFINITY RULES

$$v = (V_{max} \cdot [S]) / (K_m + [S])$$

Hexokinase (Extra-Hepatic)

- ✦ **Low Km:** High glucose affinity.
- ✦ **Basal Workhorse:** Fully saturated at normal blood glucose (ensures brain/RBC get glucose).

Glucokinase (Liver & β Cells)

- ✦ **High Km:** Low glucose affinity.
- ✦ **Glucose Sensor:** Active only after meals to store glycogen; proportional to blood glucose.

- ✦ **Km Definition:** Substrate concentration at which reaction rate is exactly half of maximum velocity ($\frac{1}{2} V_{max}$).
- ✦ **Affinity Trap:** Km and affinity are **inversely proportional**. Low Km = High Affinity; High Km = Low Affinity.

08 Isoenzymes LDH/CK ● TIER 1

 ISOENZYME REVISION CARD

LDH Distribution Pattern

LDH-1 (H₄)	Heart, RBC (High in MI)
LDH-2 (H₃M)	Normal Serum Predominant
LDH-5 (M₄)	Liver, Skeletal Muscle

Creatine Kinase (CK) Deck

CK-1 (BB)	Brain (Negligible in serum)
CK-2 (MB)	Myocardium (Cardiac MI marker)
CK-3 (MM)	Skeletal Muscle (Rises in dystrophy)

- ✦ **LDH Flip Rule:** Healthy state: **LDH-2 > LDH-1**. Acute myocardial infarction (MI) flips the profile: **LDH-1 > LDH-2** due to cardiac cell lysis.
- ✦ **CK-MB Diagnostics:** Serum level rising above **6% of total CK** indicates MI. Peaks fast (12-24 hours) — crucial early diagnostic marker.

09 Clinical Correlations ● TIER 1

 HIGH-YIELD PHARMACOLOGY PAIRS

DRUG NAME	INHIBITION TYPE	ENZYME TARGET	MEDICAL UTILITY
ATORVASTATIN	Competitive	HMG-CoA Reductase	Hyperlipidemia (lowers cholesterol)
METHOTREXATE	Competitive	Dihydrofolate Reductase	Oncology & Rheumatoid Arthritis
ALLOPURINOL	Suicide	Xanthine Oxidase	Chronic Gout (lowers uric acid)
ASPIRIN	Irreversible	Cyclooxygenase (COX-1/2)	Cardioprotection (anti-platelet)
ORGANOPHOSPHATES	Irreversible	Acetylcholinesterase	Pesticides (SLUDGE toxicity syndrome)

10 Memory Mnemonics ● TIER 3

 MNEMONIC DECK

#1 Kinetics Cheat

"Competitive Kills Km, Non-Comp Nails Vmax"

Competitive shifts Km up; Non-competitive shifts Vmax down.

#2 L-B Plot Crossings

"Competitive meets at Y, Non-Comp meets at X"

Competitive intersects on Y-axis (Vmax same). Non-competitive intersects on X-axis (Km same).

11 Essay Question Answer Template ● TIER 2



5-STEP MBBS ESSAY BLUEPRINT (MAX MARKS)

For justification questions (e.g. "Methotrexate is a competitive inhibitor, justify"):

1. Define

Define Comp. Inhibitor

2. Analog

Mimics DHFA

3. Block

Inhibits DHFR

4. Kinetics

$K_m \uparrow$, V_{max} same

5. Clinical

Folinic Acid Rescue

- ✦ **Write Flowcharts First:** Medical examiners scan flowcharts and kinetic parameters (K_m/V_{max}) first. Always sketch the L-B plot!
- ✦ **Use Clinical Rescue:** Always mention how to bypass the drug block (e.g. *Leucovorin/Folinic Acid* rescue for Methotrexate block).

12 Viva Rapid Fire ● TIER 3



VIVA CORE Q&A DECK (TOGGLE TO EXPAND)



EXAMINER FAV QUESTIONS

Why is allopurinol classified as a "suicide" inhibitor?

Allopurinol is a substrate analog that is first metabolized by Xanthine Oxidase to Alloxanthine. Alloxanthine then covalently complexes with the molybdenum core of Xanthine Oxidase at the active site, permanently and irreversibly deactivating it. The enzyme effectively commits suicide by metabolizing its own inhibitor.

How does an uncoupler differ from an ETC chain inhibitor?

An **ETC Inhibitor** (e.g., Cyanide) blocks electron flow through the complexes directly, arresting **both** electron transport and ATP synthesis. An **Uncoupler** (e.g., 2,4-DNP) does not block electron flow; instead, it collapses the proton gradient by carrying protons back across the inner membrane. Electron transport continues rapidly, but **no ATP is synthesized** — energy dissipates strictly as heat.

What is the "flipped" LDH isoenzyme pattern, and when is it seen?

Normally, serum **LDH-2 is higher than LDH-1**. However, during a **Myocardial Infarction (MI)**, heart muscle tissue undergoes necrosis, spilling huge volumes of LDH-1 (which is abundant in heart cells) into circulation. This inverts the ratio so that **LDH-1 > LDH-2**, known as the **"flipped" LDH pattern**.

MBBS Biochemistry – Module 01: Enzyme Inhibition (Revision Edition)

Compiled from PYQ 2010–2025

Designed visually for high-impact revision. Fit for A4 PDF Print.

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Version 2.0 · Pilot Print-Ready Revision Edition